

First-Year Second-Semester Exam

May 2021

Answer the following problems and show all your work. Support all statements to the best of your ability. Use a separate sheet of paper for each problem. Each is worth 15 points.

1. (a) State the Urysohn Lemma.
(b) Let X be a connected normal space having two distinct points x and y . Show that X has uncountably many points.
2. (a) State the Brouwer fixed-point theorem for the disc B^2 .
(b) Let A be a 3 by 3 matrix of positive real numbers. Show that A has a positive real eigenvalue. Hint: Let B be the intersection of the unit sphere S^2 and the positive octant, $\{(x, y, z \in \mathbb{R}^3 \mid x, y, z \geq 0\}$.
3. Prove that $\pi_1(S^1, 1) \cong \mathbb{Z}$.
4. (a) Define the Hawaiian earring space, H .
(b) Show the images of the generators of the fundamental groups of the circles do not generate the fundamental group of H .
5. (a) Define surface.
(b) Compute the fundamental group of the connected sum of a torus and second torus $T^2 \# T^2$.

Answer the following with complete definitions or statements or short proofs. Each is worth 5 points.

6. Show that $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^3$.
7. If $j : A \hookrightarrow X$ is the inclusion of a retract, then show that the induced map on fundamental groups $j_\#$ is injective.
8. State the *Seifert-van Kampen Theorem*.
9. What is the fundamental group of the torus $T^2 = S^1 \times S^1$?
10. Given continuous maps of topological spaces $f : A \rightarrow X$ and $g : A \rightarrow Y$ state the universal property satisfied by the pushout.